BANKURA UNIVERSITY

B. Sc. Semester I (Hons) Examination 2017

MATHEMATICS

Subject Code : 12102 Course Code : UG/SC/102/C-02

Course Title : Algebra

Full Marks : 40 **Time : 2 Hours**

The figures in the right hand side margin indicate marks.

1. Answer any five questions : **2x5=10**

a) If the sum of two roots of the equation $x^3+px^2+qx+r=0$ is zero, then show that $pq=r$

b) Express the complex number $-1-i$ in Polar form with Principal argument.

c) z is a variable complex number such that an amplitude of $\frac{z-i}{z+1}$ is $\pi/4$. Show that the point z lies on a circle in the complex plane.

d) Show that for any three real numbers a, b, c

$a^8+b^8+c^8 \geq a^2 b^2 c^2 (ab+bc+ca)$.

e) Let λ be an eigen value of an n x n matrix A. Show that λ^2 is an eigen value of A^2.

f) Is the following transformation linear?

$T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$T(x,y) = (x+y, 0), \forall x, y \in \mathbb{R}$

g) Suppose R and S are two equivalence relations on a nonempty set A. Verify whether $R \cup S$ and $R \cap S$ are equivalence relations on A.

h) Use the theory of congruence to find the remainder when the sum $1^3+2^3+3^3+.....+99^3$ is divided by 3.

2. Answer any four questions : **5x4=20**

a) i) If α, β, γ be the roots of the equation $x^3+px^2+qx+r=0$, find the equation whose roots are $\alpha\beta+\beta\gamma,\ \beta\gamma+\gamma\alpha,\ \gamma\alpha+\alpha\beta$. 3

ii) Find the minimum number of complex roots of the equation $x^7-3x^3+x^2=0$. 2

b) Determine the values of a and b for which the system 2+2+1

$$x+y+z=1$$
$$x+2y-z=b$$
$$5x+7y+az=b^2$$

has no solution, only one solution, many solutions respectively.

c) If

$$A=\begin{pmatrix}1 & -3 & 3\\ 3 & -5 & 3\\ 6 & -6 & 4\end{pmatrix},$$

then find three eigen vectors of A which are linearly independent. 5

d) i) If a is prime to b then prove that a^2+b^2 is prime to a^2b^2. 2

ii) Prove that the product of k consecutive integers is divisible by k. 3

e) A relation β is defined on $\mathbb{Z}$ by "$x\ \beta\ y$ if and only if x^2-y^2 is divisible by 5" for $x, y, \in \mathbb{Z}$. Prove that β is an equivalence relation on $\mathbb{Z}$ and find the distinct equivalence classes. 5

f) Let V(F) be a vector space and $S \neq \phi$ be a finite subset of V. Then prove the set W of all linear combinations of the elements of S forms a subspace of V. Also show that W is the smallest subspace of V containing S. 3+2

3. Answer either (a) or (b) : 10 x 1 = 10

a) i) Show that the sets $A=\{x\in\mathbb{R} : 0\le x\le 1\}$ and $B=\{x\in\mathbb{R} : a\le x\le b\}$ have same cardinal number. 3

ii) If $|z|=1$ and amp $z=\theta$, $(0<\theta<\pi)$ then find the modulus and principal amplitude of $\frac{2}{1+z}$. 4

iii) Prove that for $n>3$, the integers n, $n+2$, $n+4$ cannot be all primes. 3

b) i) State and prove the Cauchy - Schwarz inequality. 6

ii) Verify Caley - Hamilton theorem for the matrix 4

$$A=\begin{pmatrix}0 & 0 & 1\\ 3 & 1 & 0\\ -2 & 1 & 4\end{pmatrix}.$$ Hence find A^{-1}.

B.Sc. Semester I (Honours) Examination, 2018-19
MATHEMATICS

Course Id : 12112 **Course Code : SHMTH-102C-2(T)**

Course Title : Algebra

Time: 2 Hours **Full Marks: 40**

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

1. Answer *any five* questions: 2×5=10

(a) Use De Moivre's theorem to prove that $\tan 4\theta = \frac{4\tan\theta - 4\tan^3\theta}{1-6\tan^2\theta+\tan^4\theta}$.

(b) State Division algorithm. Use it to find the remainder when – 326 is divided by 5. 1+1=2

(c) If $S = a+b+c$, prove that $\frac{s}{a-b} + \frac{s}{s-b} + \frac{s}{s-c} > {}^{9}/_{2}$.

(d) Find all the equivalence relations on the set $S = \{a, b, c\}$.

(e) For what values of k, the following system of equations has a non-trivial solution?

$$x + 2y + 3z = kx, 2x + y + 3z = ky, 2x + 3y + z = kz$$

(f) When a transformation $T: R^m \to R^n$ is said to be a linear transformation? Is the following transformation linear?

$T: R^2 \to R$ defined by

$T(x, y) = |x - y|$, for all $x, y \in R$.

(g) If an equation $f(x) = 0$ with real coefficients consists of only even powers of x with all positive signs, show with proper reason that the equation cannot have a real root.

(h) Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -2 & 5 \\ 0 & 0 & 2 \end{pmatrix}$. Find the eigenvalues of the matrix $A^5 - I_3$.

2. Answer *any four* questions: 5×4=20

(a) Define rank of a matrix. Find the row reduced echelon form of the matrix

$\begin{pmatrix} 1 & -1 & 2 & 0 & 4 \\ 2 & 2 & 1 & 5 & 2 \\ 1 & 3 & -1 & 0 & 3 \\ 1 & 7 & -4 & 1 & 1 \end{pmatrix}$ and find its rank. 1+3+1=5

(b) (i) If $ax \equiv ay (mod\ n)$ and a is prime to n, then show that $x \equiv y (mod\ n)$.

(ii) Obtain the equation whose roots are the roots of the equation $x^4 - 8x^2 + 8x + 6 = 0$, each diminished by 2. 1+4=5

(c) If a, b, c be positive real numbers such that the sum of any two is greater than the third, then prove that $abc \geq (a+b-c)(b+c-a)(c+a-b)$. Derive when equality occurs. 4+1=5

(d) State the second principle of mathematical induction and using this principle, prove that $(3+\sqrt{5})^n + (3-\sqrt{5})^n$ is divisible by 2^n, for all $n \in N$ (*N* be set of natural numbers). 5

(e) Show that the eigenvalues of a real symmetric matrix are all real. 5

(f) (i) Show that $A = \{(x_1, x_2, x_3, \ldots x_n) \in R^n : x_1 + x_2 + \cdots + x_n = 0\}$ is a subspace of R^n. Find the dimension of the subspace A of R^n.

(ii) Is the union of two subspaces of R^n, a subspace of R^n? Justify your answer. 3+2=5

3. Answer *any one* question: 10×1=10

(a) (i) Show that the transformation $T: R^3 \to R^2$ defined by $T(x, y, z) = (z, x+y)$ is linear.

(ii) Let $f: A \to B$ and $g: B \to C$ be two mappings such that $g_\circ f: A \to C$ is injective. Is it necessary that g is injective? Justify your answer.

(iii) Solve the cubic equation $x^3 - 9x + 28 = 0$ by Cardon's method. 3+2+5=10

(b) (i) Prove that any partition of a non-empty set S induces an equivalence relation on S.

(ii) Prove or disprove:

"Composition of two linear transformation from $R^m \to R^n$ is linear transformation."

(iii) A linear mapping $T: \mathbb{R}^3 \to \mathbb{R}^3$ is defined by $T(x_1, x_2, x_3) = (2x_1 + x_2 - x_3, x_2 + 4x_3, x_1 - x_2 + 3x_3)(x_1, x_2, x_3) \in R^3$. Obtain the matrix of *T* relative to the ordered base (0, 1, 1), (1, 0, 1), (1, 1, 0) of $\mathbb{R}^3$. 3+2+5=10

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SH-I/Mathematics-102/C-2/19

B.Sc. 1st Semester (Honours) Examination, 2019-20

MATHEMATICS

Course ID : 12112 **Course Code : SH/MTH/102/C-2**

Course Title : Algebra

Time 2 Hours **Full Marks: 40**

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Unless otherwise mentioned, notations and symbols have their usual meaning.

1. Answer *any five* questions: 2×5=10

(a) Is union of two equivalence relations an equivalence relation? (Justify).

(b) If x, y, z are three positive real numbers, show that $\frac{yz}{x}+\frac{xz}{y}+\frac{xy}{z} \geq x+y+z$.

(c) If A be an invertible matrix, then show that A^{-1} is invertible and $(A^{-1})^{-1}=A$.

(d) A matrix A has eigenvalues 1 and 4 with corresponding eigenvectors $(1,-1)^T$ and $(2,1)^T$ respectively. Find the matrix A.

(e) Prove that $\sqrt[n]{i}+\sqrt[n]{-i}=2cos\frac{\pi}{2n}$.

(f) Find the remainder when 4^{119} is divided by 9.

(g) V and W are two subspaces of R^n and $T : V \rightarrow W$ is a linear transformation. Prove that $T(\theta_v)=\theta_{\mathrm{w}}$ where the symbols have the usual meaning. Hence show that

$T(-\alpha)=-T(\alpha)\ \forall\ \alpha\ \in\ V.$

(h) If α, β, γ are the roots of $x^3-px^2+qx-r=0$, obtain the value of $\sum\frac{1}{\alpha^2}$.

2. Answer *any four* questions: 5×4=20

(a) (i) A relation ρ on Z, (Z, be the set of integers), such that $x\rho y$ if and only if $4x+7y$ is divisible 11, for $x, y\ \in\ Z$. Verify the relation ρ is an equivalence relation on Z or not.

(ii) $n(>1)$ be a positive integer then show that $(n+1)^{n-1}\ (n+2)^n > 3^n(n!)^2$. 3+2=5

(b) (i) Define cardinality of a set. Do the sets Z and N have same cardinal number? (Z and N be the set of integers and natural numbers.)

(ii) If α be a root of the equation $x^7=1$, then show that the roots of the equation.

$x^2+x+2=0$ are $(\alpha+\alpha^2+\alpha^4)$ and $(\alpha^3+\alpha^5+\alpha^6)$. (1+2)+2=5

(c) (i) Find the greatest eigen value and the corresponding eigen vector of the matrix

$$A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$$

(ii) Using mathematical induction, prove that there are 2^n subsets of a set of n elements. 3+2=5

(d) Let $f: R \to R$ be a mapping defined by

$f(x) = |\, x \,| + x, x \in R$ and

$g : R \to R$ be another mapping defined by $g(x) = |\, x \,| - x,\ x \in R$;

Find the compositions $g \circ f$ and $f \circ g$. 5

(e) Determine the linear mapping $T : R^3 \to R^3$ that maps the basis vectors (0, 1, 1), (1, 0, 1), (1, 1, 0) of R^3 to (2, 1, 1), (1, 2, 1), (1, 1, 2) respectively. 5

(f) The eigenvalues of a 3×3 matrix A are in A.P. and given that $|A| = 80$, Trace $A = 15$. Find the eigenvalues. 5

3. Answer *any one* question: 10×1=10

(a) (i) Examine if the relation ρ defined on Z by $\rho = \{(a, b) \in Z \times Z : 7/3a + 4b\}$ is an equivalence relation.

(ii) Verify Caley-Hamilton theorem for the square matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$ Hence find A^{-1}.

(iii) From the result of 3a (ii) above, compute A^{100}. 3+4+3=10

(b) (i) If $cl(a)$ is an equivalence class and $b \in cl(a)$ then prove that $cl(a) = cl(b)$.

(ii) Find $g \circ f$ and $f \circ g$ if $f : R \to R$ be defined by $f(x) = |x| + x, x \in R$ g: and $R \to R$ be defined by $g(x) = |\, x \,| - x,\ x \in R$.

(iii) Determine the condition for which the following system of equation has

(I) only one solution (II) no solution (III) many solution

$$x + y + z = b$$
$$2x + y + 3z = b + 1$$
$$5x + 2y + az = b^2$$

2+3+5=10

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B.SC. FIRST SEMESTER (HONS.) EXAMINATION 2021

Subject: Mathematics **Course ID: 12112**

Course Title: Algebra **Course code: SH/MTH/102/C-2**

Full Marks: 40 **Time: 2 Hours**

The figures in the margin indicate full marks

Notations and symbols have their usual meaning.

1. Answer any five questions: $2 \times 5 = 10$

(a) Find the product of all the values of $(1+\sqrt{3}i)^{\frac{3}{4}}$.

(b) Prove that $3^{2n-1} + 2^{n+1}$ is divisible by 7, using principle of mathematical induction.

(c) Show that the set $S = \{(x,y,z) : x,y,z \in \mathbb{R}, 2x+y-z = 0\}$ is a subspace of $\mathbb{R}^3$.

(d) Use division algorithm to find integer u and v satisfying $30u + 72v = 12$.

(e) Let $f: \mathbb{Z} \to \mathbb{Z}$ be a mapping defined by $f(x) = x^2, x \in \mathbb{Z}$ and $g: \mathbb{Z} \to \mathbb{Z}$ be a mapping defined by $g(x) = 2x, x \in \mathbb{Z}$. Find fog and gof.

(f) If A be a set of nonzero integers and R be a relation defined on $A \times A$ such that $(a,b)R(c,d)$ iff $ad = bc$, then show that R is an equivalence relation.

(g) If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, then find the equation whose roots are $(\alpha+\beta)/\gamma$, $(\beta+\gamma)/\alpha, (\gamma+\alpha)/\beta$.

(h) Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be defined by $T(x_1,x_2,x_3) = (x_1+1, x_2+1, x_3+1)$. Examine whether T is a linear mapping or not.

2. Answer any four questions: $5 \times 4 = 20$

(a) (i) Show that if $n \geq 4$, then $n, n+2$ and $n+4$ cannot be all prime.

(ii) Use theory of congruence to find the remainder when $1! + 2! + 3! + + \cdots + 100!$ is divided by 15. 3+2=5

(b) (i) Use Cauchy- Schwarz inequality to prove that $(a+b+c+d)(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}) > 16$, where a, b, c, d are positive real numbers, not all equal.

(ii) If x, y, z are positive real numbers and $x+y+z = 1$, prove that

$8xyz \leq (1-x)(1-y)(1-z) \leq \frac{8}{27}$. 2+3=5

(c) Using De Moivre's theorem, prove that $\tan 5\theta = \frac{5\tan\theta - 10\tan^3\theta + \tan^5\theta}{1 - 10\tan^2\theta + 5\tan^4\theta}$

(d) Solve the equation $x^4 - 6x^2 + 16x - 15 = 0$ by Ferrari's method.

(e) Determine the conditions for which the following system of equations has

(I) only one solution (II) no solution (III) many solutions

$$x + y + z = 1$$
$$x + 2y - z = b$$
$$5x + 7y + az = b^2. \qquad 5$$

(f) Verify Cayley-Hamilton theorem for the matrix $A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 1 \\ 2 & 3 & 2 \end{pmatrix}$. Hence compute A^{-1}. 5

3. Answer any one question: $10 \times 1 = 10$

(a) (i) Apply Descarte's rule of sign to determine the nature of the roots of the equation $x^4 + 2x^2 + 3x - 1 = 0$.

(ii) Using principle of mathematical induction, prove that $5^n - 1$ is divisible by 4 for all positive integer n.

(iii) Find all eigen values of the matrix $A = \begin{pmatrix} 1 & -1 & 2 \\ 2 & -2 & 4 \\ 3 & -3 & 6 \end{pmatrix}$. Hence find eigen vector corresponding to the smallest eigen value. 2+3+(2+3)=10

(b) (i) If α, β, γ are the roots of the equation $x^3 + qx + r = 0$ $(r \neq 0)$, find the equation whose roots are $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}, \frac{\beta}{\gamma} + \frac{\gamma}{\beta}, \frac{\gamma}{\alpha} + \frac{\alpha}{\gamma}$.

(ii) Find $mod\ z$ and $amp\ z$ when $z = (1 + i)^7$.

(iii) Determine the representative matrix of the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (3x + z, -2x + y, -x + 2y + 4z)$, w.r.t. the standard basis of $\mathbb{R}^3$. 3+4+3=10

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B.SC. FIRST SEMESTER (HONOURS) EXAMINATIONS, 2022

Subject: Mathematics **Course ID: 12112**

Course Code: SH/MTH/102/C-2

Course Title: Algebra

Full Marks: 40 **Time: 2 Hours**

The figures in the margin indicate full marks

Notations and symbols have their usual meaning.

1. Answer any five questions: **2 × 5 = 10**

(a) Let the roots of the equation $x^3 + px^2 + qx + r = 0$ be α, β, γ. Find the equation whose roots are

$\alpha - \frac{\beta\gamma}{\alpha}$, $\beta - \frac{\gamma\alpha}{\beta}$ and $\gamma - \frac{\alpha\beta}{\gamma}$.

(b) If $\{\alpha, \beta, \gamma\}$ forms abasis of a subspace, then show that $\{\alpha + \beta + \gamma, \beta + \gamma, \gamma\}$ also forms a basis of that subspace.

(c) Determine the eigen values of the matrix A=$\begin{pmatrix} a & h & g \\ 0 & b & 0 \\ 0 & c & c \end{pmatrix}$.

(d) Let a function $f: \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \begin{cases} 3x - 1 & for\ x > 3 \\ 2x^2 & for - 2 < x \leq 3 \\ 3x^2 - 7 & for\ x \leq -2. \end{cases}$

Find $f^{-1}(5)$.

(e) If f, g, h are three functions from $\mathbb{Z}\ to\ \mathbb{Z}$, such that $(z) = n^2$, $g(n) = n + 1$ and $h(n) = n - 1$. Find $g \circ f \circ h$, $f \circ g \circ h$ and $h \circ f \circ g$.

(f) Solve the equation $x^3 - 7x^2 + 36 = 0$, given that one of its roots is double of another.

(g) Find the smallest positive residue in $2^{41} (mod\ 23)$.

(h) If a, b, c are any three integers such that $gcd(a, c) = 1$ and $gcd(b, c) = 1$, then show that $gcd(ab, c) = 1$.

2. Answer any four questions: **5× 4 = 20**

(a) Find all the values of $(i)^{1/2} + (-i)^{1/2}$.

(b) If α, β, γ are the roots of the equation$x^3 - x^2 + ax + b = 0$ and β, γ, δ are the roots of the equation $x^3 - 4x^2 + mx + n = 0$, and also $\alpha, \beta, \gamma, \delta$ are in A.P. , then show that

$$b = m + n - 3.$$

(c) If a, b, c are three positive real numbers such that $abc = 1$, then prove that

$$\frac{1 + ab}{1 + a} + \frac{1 + bc}{1 + b} + \frac{1 + ca}{1 + c} \geq 3$$

(d) Prove that the product of any three consecutive integers is divisible by 6.

(e) Prove by using Mathematical Induction that $3^{2n} - 8n - 1$ is divisible by 64.

(f) Find all real values of z for which the rank of the matrix $\begin{pmatrix} 1+z & 2 & 3 & 4 \\ 1 & 2+z & 3 & 4 \\ 1 & 2 & 3+z & 4 \\ 1 & 2 & 3 & 4+z \end{pmatrix}$

is less than 4.

3. _Answer any one question:_ **10 x 1 = 10**

(a) (i) Find the general solutions of $\sin z = 2i$.

(ii) Find the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ which transforms the basis vectors $(1,2)$ and $(0,1)$ to $(3,-1,5)$ and $(2,1,-1)$.

(iii) Obtain the characteristic equation of the matrix

$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{pmatrix}$ and verify that A satisfies this equation. Hence find A^{-1}.

(3+2)+5

(b)(i) Express $(1+i)^{-i}$ in the form of $A + iB$.

(ii) For which values of 'k' the following system of equations have a solution?

$$x + y + z = 1, x + 2y + 4z = k, x + 4y + 10z = k^2$$

and solve them completely for each value of 'k'.

5+5

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B.Sc. 1st Semester (Honours) Examination-2022-23

MATHEMATICS

Course ID : 12112 Course Code : SH/MTH/102/C-2

Course Title : Algebra (New)

Time : 2 Hours *Full Marks* : 40

The figures in the right hand margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Notations and symbols have their usual meaning.

Unit–I

1. Answer **any *five*** questions : 2×5=10

(a) Find the product of all the values of $\left(\sqrt{3}+i\right)^{\frac{3}{5}}$.

(b) Find the dimension of the Subspace S of $\mathbb{R}^4$ defined by $S=\{(x,y,z,w)\in\mathbb{R}^4 : x+2y-z=0,\ 2x+y+w=0\}$.

(c) Use Euclidean algorithm to find two integers u and v such that gcd $(13,80)=13u+80v$.

(d) Determine a Linear mapping $T:R^3\to R^3$ which maps the basis vectors $\{(0,1,1), (1,0,1)$ and $(1,1,0)\}$ to the vectors $\{(2,1,1), (1,2,1)$ and $(1,1,2)\}$ respectively.

(Turn Over)

(e) If α be an eigen value of an $n \times n$ idempotent matrix A, then show that α is either 0 or 1.

(f) examine if the relation p defined on $N \times N$ by "$(a,b)p(c,d)$ holds if and only if $ad = bc$" is reflexive, symmetric and transitive or not.

(g) Find the rank of the matrix $\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 4 & 4 & 4 & 8 & 0 \\ 8 & 2 & 0 & 10 & 2 \\ 6 & 3 & 2 & 9 & 1 \end{pmatrix}$

(h) Find the number of real positive and negative roots of the equation $x^3-7x+7=0$, using Strum's theorem.

Unit–II

2. Answer any *four* questions : 5×4=20

(a) (i) Show that the product of any m consecutive integers is divisible by m.

(ii) Let $f,g:\mathbb{R}\to\mathbb{R}$ be defined by $f(x)=3x^2-5$ and $g(x)=\frac{x}{x^2+1}$. Find fog and gof. Also show that $g^{-1}(g(2,3\}))\neq\{2,3\}$. 2+3

(b) Determine the condition for which the system of equations has (i) unique solution (ii) no solution and (iii) many solutions :

$x + 2y + z = 1,\ 2x + y + 3z = b,\ x + ay + 3z = b + 1$

(c) (i) Find arg z where $z = 1 + i\tan\frac{3\pi}{5}$.

(ii) In n be a positive integer and $(1 + z)^n = p_0 + p_1 z + p_2 z^2 + \cdots + p_n z^n$. Prove that $p_0 - p_2 + p_4 - \cdots = 2^{n/2}\cos\frac{n\pi}{4}$ and $p_1 - p_3 - p_5 - \cdots = 2^{n/2}\sin\frac{n\pi}{4}$.

2+3

(d) (i) Let α, β, γ be the roots of the equation $x^3 + ax^2 + bx + c = 0$. Find the value of $\Sigma\alpha^2\beta^2$.

(ii) Solve the equation by Cardan's method : $x^3 - 27x - 54 = 0$. 2+3

(e) (i) Let A be a 3×3 matrix with eigen values are $1, -1, 0$. Find all eigen values of the matrix $I + A^{2022}$.

(ii) Find the algebraic and the geometric multiplicities of each eigen values of the matrix

$$\begin{pmatrix} 1 & -1 & 0 \\ 1 & 2 & -1 \\ 3 & 2 & -2 \end{pmatrix}$$

1+4

(f) (i) If x, y, z be three unequal positive numbers, then show that $\left(\frac{y+z}{2}\right)^x \left(\frac{z+x}{2}\right)^y \left(\frac{x+y}{2}\right)^z < x^x y^y z^z$.

(ii) Show that $2^{73} + 14^3 \equiv 2 \mod (11)$ 3+2

Unit–III

3. Answer **any** ***one*** question : 10×1=10

(a) (i) State Caley–Hamilton theorem for a matrix A. Using this theorem find A^{50}, where $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

(ii) Show that every integer and its cube leave the same remainder when divided by 6.

(iii) Using the principle of Mathematical Induction, prove that $7^{2n} + 16n - 1$ is divisible by 64, for all $n \in N$. (1+3)+3+3

(b) (i) Apply Descarts rule of signs to find the nature of the roots of the equation $x^4 + 2x^4 + 3x - 1 = 0$.

(ii) Use principle of induction to prove that $2 \cdot 7^n + 3 \cdot 5^n - 5$ is divisible by 24 for all $n \in \mathbb{N}$.

(iii) Let the matrix of a linear mapping $T : \mathbb{R}^3 \to \mathbb{R}^3$ relative to the ordered bases ((0,1,1), (1,0,1), (1,1,0)) of $\mathbb{R}^3$ be $\begin{pmatrix} 0 & 3 & 0 \\ 2 & 3 & -2 \\ 2 & -1 & 2 \end{pmatrix}$. Find T. Find the matrix of T relative to the ordered basis (2,1,1), (1,2,1), (1,1,2) of $\mathbb{R}^3$. 2+3+5

Course Title : Algebra (Old)

Unit–I

1. Answer any *five* question : 2×5=10

(a) Find the remainder when 11^{40} is divided by 8.

(b) Let a,b,c be all positive real numbers. Then prove that

$$\frac{a^2+b^2}{a+b}+\frac{b^2+c^2}{b+c}+\frac{c^2+a^2}{c+a}\geq a+b+c.$$

(c) Let $f:\mathbb{Z}\to E_+^0$ be defined by $f(x)=|x|+x \;\; \forall x\in\mathbb{Z}$ *and* E_+^0 be the set of all nonnegative even integers. Let $g:E_+^0\to\mathbb{Z}$ be defined by $g(x)=\frac{x}{2} \;\; \forall x\in E_+^0$.

Check whether g is inverse of f.

(d) Check whether $f:\mathbb{Z}\to\mathbb{Z}$ defined by $f(n)=4n-5$ $\forall n\in\mathbb{Z}$ is bijective.

(e) If $a_1, a_2, \ldots, b_1, b_2, \ldots, b_n; c_1, c_2, \ldots, c_n$ are all positive real numbers then prove that $(a_1 b_1 c_1 + a_2 b_2 c_2 + \cdots + a_n b_n c_n)^2 < (a_1^2+a_2^2+\cdots+a_n^2)(b_1^2+b_2^2+\cdots+b_n^2)(c_1^2+c_2^2+\cdots+c_n^2)$.

(f) Solve the equation $x^6+x^5+x^4+x^3+x^2+x+1=0$.

(g) Apply Descartes' rule of signs to examine the nature of the roots of the equation $x^4 + 2x^2 + 3x - 1 = 0$.

(h) Use Cayley–Hamilton theorem to compute A^{-1} where

$$A = \begin{pmatrix} 2 & 1 \\ 3 & 5 \end{pmatrix}.$$

Unit–II

2. Answer **any *four*** questions : 5×4=20

(a) (i) Let $f : A \to B$ and $g : B \to C$ be both bijections. Prove that $g \circ f : A \to C$ is invertible and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. 3

(ii) Suppose f and g are two functions from $\mathbb{R}$ into $\mathbb{R}$ such that $f \circ g = g \circ f$. Does it necessarily imply that $f = g$? Justify your answer. 2

(b) (i) Find gcd of 615 and 1080 and find integers s and t such that gcd $(615, 1080) = 615\, s + 1080\, t$. 2

(ii) What is the remainder when $1! + 2! + 3! + \cdots + 99! + 100!$ is divided by 18? 3

(c) A relation ρ is defined on $\mathbb{Z}$ by "$x \,\rho\, y$ if and only if $x^2 - y^2$ is divisible by 5" for $x, y, \in \mathbb{Z}$. Prove that ρ is an equivalence relation on $\mathbb{Z}$. Show that there are three distinct equivalence classes. 5

(d) (i) Find the principal value of $(1 + i)^i$. 2

(ii) Find the general solution of $\cos z = 2$. 3

(e) Find the equation whose roots are the roots of $x^4 - 8x^2 + 8x + 6 = 0$, each diminished by 2. 5

(f) Show that eigen values of a real symmetric matrix are all real. 5

Unit–III

3. Answer **any *one*** question : 10×1=10

(a) (i) If $A = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix}$, show that $A^2 - 10A + 16I_3 = 0$.

Hence obtain A^{-1}. 3

(ii) Find all real x for which the rank of the matrix is less than 4. 2

$$\begin{pmatrix} x & 1 & 1 & 1 \\ 1 & x & 1 & 1 \\ 1 & 1 & x & 1 \\ 1 & 1 & 1 & x \end{pmatrix}$$

(iii) Determine the conditions for which the system

$$x + y + z = 1$$
$$x + 2y - z = b$$
$$5x + 7y + az = b^2$$

admits of (i) only one solution, (ii) no solution, (iii) infinitely many solution. 5

(b) (i) A linear mapping $T: \mathbb{R}^3 \to \mathbb{R}^3$ is defined by

$$T(x_1, x_2, x_3) = (2x_1 + x_2 - x_3,\ x^2 + 4x_3,\ x_1 - x_2 + 3x_3),$$

$(x_1, x_2, x_3) \in \mathbb{R}^3$.

Find the matrix to T relative to the ordered basis ((0,0,1), (1,0,0), (0,1,0)) of $\mathbb{R}^3$. 3

(ii) Check whether the relations p on $\mathbb{Z}$ defined by "$x\ p\ y$ *if f* $x | y$" is antisymmetric. 2

(iii) Find whether the following is bijective; if so, find their inverse.

Let $S = \{x \in \mathbb{R} : -1 < x < 1\}$ and $f: \mathbb{R} \to S$ defined by

$$f(x) = \frac{x}{1+|x|}.$$ 5

B.Sc. 1st Semester (Honours) Examination, 2023-24

MATHEMATICS

Course ID : 12112 Course Code : SH-MTH-102/C-2

Course Title : Algebra

[Old Syllabus-2017]

Time : 2 Hours Full Marks : 40

The figures in the right margin indicate marks.

Candidates are required to answer in their own words as far as practicable.

UNIT-I

1. Answer *any five* from the following questions: 5×2=10

a) Show that $-1, i, -i, \frac{1}{2}(\sqrt{2}+i\sqrt{2})$ are concyclic.

b) Prove that $2^n > 1+\sqrt{2^{n-1}}\,n$.

c) How many symmetric relations can be defined on a set with n elements?

d) Find all the real values of x for which the rank of the matrix A is 2, where

$$A=\begin{bmatrix} 2 & 1 & 4 \\ 1 & x & 2 \\ 4 & 0 & x+2 \end{bmatrix}.$$

e) Show that for any three real number a, b, c

$$a^8+b^8+c^8 \geq a^2b^2c^2(ab+bc+ca).$$

f) Find mod Z and amp Z when $Z = (1+i)^7$.

g) Prove that $3^{2n-1} + 2^{n+1}$ is divisible by 7, using principle of mathematical induction.

h) Show that the set

$S = \{(x, y, z) : x, y, z \in \mathbb{R}, 2x + y - z = 0\}$ is a subspace of $\mathbb{R}^3$.

UNIT-II

2. Answer ***any four*** from the following questions:

4×5=20

a) If a, b, c are three positive numbers and a+b+c = 1, then prove that,

$$8abc \le (1-a)(1-b)(1-c) \le \frac{8}{27}.$$

b) Let α, β, γ be the roots of the equation, $x^3 + px^2 + qx + r = 0$, Then find the value of

$$\left(\frac{1}{\beta}+\frac{1}{\gamma}-\frac{1}{\alpha}\right)\left(\frac{1}{\gamma}+\frac{1}{\alpha}-\frac{1}{\beta}\right)\left(\frac{1}{\alpha}+\frac{1}{\beta}-\frac{1}{\gamma}\right).$$

c) Determine the conditions for which the system of equations has (a) only one solution, (b) no solution, (c) many solutions.

$$x + y + z = b,\ 2x + y + 3z = b + 1,\ 5x + 2y + az = b^2.$$

d) Find the condition for the equation $X^3 + 3HX + G = 0$ to have three distinct real roots.

e) Stat and Prove the Cauchy-Schwarz inequality.

f) Define rank of a matrix. find the row reduced echelon form of the matrix.

$$\begin{pmatrix} 1 & -1 & 2 & 0 & 4 \\ 2 & 2 & 1 & 5 & 2 \\ 1 & 3 & -1 & 0 & 3 \\ 1 & 7 & -4 & 1 & 1 \end{pmatrix}$$ and find its rank.

UNIT-III

3. Answer ***any one*** from the following questions:

$1\times10=10$

a) i) Solve the euqation $x^3 - 30x + 133 = 0$, Using cardan's method.

ii) Determine the eigen values and eigen vector of the matrix $\begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$

iii) Use Cauchy-Schwarz inequality to prove that $(a+b+c+d)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right) > 16$, where a, b, c, d are positive real numbers not all equal. 4+4+2

b) i) Show that $T: R^3 \to R^3$ defined by $T(x, y, z) = (x + y + z, 2x + y + 2z, x + 2y + z)$ is a linear transformation. Find its kernel and image space.

ii) Show that one of the values of

$$(1 + i\sqrt{3})^{\frac{3}{4}} + (1 - i\sqrt{3})^{\frac{3}{4}} \text{ is } \sqrt[4]{32}$$

iii) Prove that $(n^2 + 2)$ is not divisible by 4 for any integer n. 4+3+3

B.Sc. 1st Semester (Honours) Examination, 2023-24

MATHEMATICS

Course ID : 12112 **Course Code : SH-MTH-102/C-2**

Course Title : Algebra

[New Syllabus-2022]

Time : 2 Hours Full Marks : 40

The figures in the right margin indicate marks.

Candidates are required to answer in their own words as far as practicable.

Notations and symbols have their usual meaning.

UNIT-I

1. Answer ***any five*** from the following questions: 5×2=10

a) Let $a_1, a_2, \ldots, a_n; b_1, b_2, \ldots, b_n$ and $c_1, c_2, \ldots, c_n$ be all positive real numbers. Apply Cauchy-Schwarz inequality to prove that $(a_1b_1c_1 + a_2b_2c_2 + \cdots + a_nb_nc_n)^2 <$

$$(a_1{}^2 + a_2{}^2 + \cdots + a_n{}^2)(b_1{}^2 + b_2{}^2 + \cdots + b_n^2)\,(c_1{}^2 + c_2{}^2 + \cdots + c_n{}^2)$$

b) Prove that for a complex number $z, |z| \geq \frac{1}{\sqrt{2}}(|Rez| + |Imz|)$.

c) If A and B are two invertible matrices, then show that AB is also invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

[Turn Over]

d) Prove that product of any three consecutive integers is divisible by 6.

e) Let W be the subspace of $\mathbb{R}^3$ defined by

$$W = \{(x, y, z) \in \mathbb{R}^3 : 2x + 2y + z = 0, 3x + 3y - z = 0, x + y - 3z = 0\}.$$

Find the dimension of W.

f) Solve for x and y, where $x, y \in Z$ and $35x + 40y = 50$.

g) The eigen values of the matrix A are 5 and 3. Find the eigen values of the matrix $(A^2 - 8A + 15I + 30A^{-1})$.

h) Let $X = \{a, b, c\}$ be a non-empty set and $P(X)$ be the power set of X. Show that $(P(X), \leq)$ is a poset.

UNIT-II

2. Answer *any four* from the following questions: $4 \times 5 = 20$

a) i) Let $\rho = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a - b$ is a multiple of $6\}$. Show that ρ is an equivalence relation.

ii) Apply Descartes' rule of signs to examine the nature of the roots of the equation $x^6 - 3x^2 - 2x - 3 = 0$.

b) i) Use Principle of mathematical induction to prove that $2^{5n+1}+3^{2n+1}$ is divisible by 23 for $n \in \mathbb{N}$.

ii) Let $f, g : \mathbb{R} \to \mathbb{R}$ be two functions, given by $f(x) = |x| + x$ for all $x \in \mathbb{R}$ and $g(x) = |x| - x$ for all $x \in \mathbb{R}$. Find fog and gof.

c) Prove that the square of any integer is always in the form of either 3n or 3n+1.

d) If x, y, z be three unequal positive numbers, then show that $\left(\frac{y+z}{2}\right)^x \left(\frac{z+x}{2}\right)^y \left(\frac{x+y}{2}\right)^z < x^x y^y z^z$.

e) Solve the equation by Ferrari's Method $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$.

f) Solve, if possible the system of linear equations

$$x_1 + 2x_2 - x_3 = 10$$
$$-x_1 + x_2 + 2x_3 = 2$$
$$2x_1 + x_2 - 3x_3 = 8.$$

UNIT-III

3. Answer *any one* from the following questions:

$1 \times 10 = 10$

a) i) Use Cayley-Hamilton theorem to find A^{100} where

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

ii) Use the theory of Congruence to find the remainder when 3.4^{n+1} is divisible by 9 for all $n \in \mathbb{N}$.

iii) Let $f : S \to \mathbb{R}$ defined by $f(x) = \frac{x}{1-|x|}, x \in S$, where $S = \{x \in \mathbb{R} : -1 < x < 1\}$. Show that the mapping f is a bijection. Determine f^{-1}.

3+3+4

b) i) Using the Principle of MAthematical Induction, prove that $7^{2n} + 16n - 1$ is divisible by 64, for all $n \in N$..

ii) Solve the equation $x^4 - 2x^2 + 8x - 3 = 0$, using Euler's method.

5+5

B. Sc. Semester II Examinations, 2023-24

Subject: Mathematics

Course ID: 22116 **Course Code: S/MTH/202/MN-2**

Course Title: Algebra [New Syllabus, NEP]

Time: 2 Hours. **Full Marks: 40**

The figures in the right-hand margin indicate marks.
Candidates are required to give their answer in their own words as far as practicable.

Notations and Symbols have their usual meaning.

Answer all the questions.

UNIT I

1. Answer any 5 (five) of the following questions:

$(2 \times 5 = 10)$

a) Show that $2^{2n+1} > 1 + (2n+1)2^n$, where n is natural number.

b) If the ratio $\frac{z-i}{z-1}$ is purely imaginary then show that the point z lies on the circle whose centre is at the point $\frac{1}{2}(1+i)$ and radius is $\frac{1}{\sqrt{2}}$.

c) If α, β, γ be the roots of $x^3 + px^2 + qx + r = 0$, find the value of $\sum \alpha^2 \beta$.

[Turn Over]

d) Let $X = \{a, b, c\}$ be a non-empty set and $P(X)$ be the power set of X. Show that $(P(X), \leq)$ is a poset, where $A \leq B$ if and only if A is a subset of B.

e) Find the rank of the matrix A, where $A = \begin{bmatrix} 3 & 5 & 7 \\ 2 & 1 & 3 \\ 1 & 4 & 4 \end{bmatrix}$.

f) Show that the eigen values of a diagonal matrix are the diagonal elements.

g) Show that $n(n+1)(n+2)$ is always divisible by 6, where n is positive integer.

h) Verify if the mapping $f: \mathbb{R}^3 \to \mathbb{R}^2$ defined by $f(x, y, z) = (3x - 2y + z, x - 3y - 2z)$ is a linear mapping or not.

UNIT II

2. Answer any 4 (four) of the following questions:

$(5 \times 4 = 20)$

a) (i) Show that the union of two equivalence relations is not necessarily an equivalence relation.

(ii) Determine the value of z when $z^6 = \sqrt{3} + i$. **[3 + 2]**

b) (i) Find the number and position of real roots of the equation $x^5 - 5x + 1 = 0$.

(ii) State the Descartes's rule of sign. **[3 + 2]**

c) State the Fundamental theorem of Algebra and show that every algebraic equation of degree 5 has 5 roots and no more.

[2 + 3]

d) (i) Solve $35x + 40y = 5$, for $x, y \in \mathbb{Z}$.

(ii) State the First Principle of mathematical induction.

[3 + 2]

e) State Cauchy-Schwarz inequality and apply it to prove that

$$(a_1b_1c_1 + a_2b_2c_2 + \cdots + a_nb_nc_n)^2$$
$$< (a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2)(c_1^2 + c_2^2 + \cdots + c_n^2).$$

f) Find the characteristic roots of matrix and verify Cayley Hamilton theorem and hence find A^{-1}, where $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$.

UNIT III

3. Answer any 1 (one) of the following questions:

$(10 \times 1 = 10)$

a) (i) Solve $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$, by Ferrari's method.

(ii) A relation R on Z (set of all integers) is defined by R = {(a, b)∈ Z x Z: a – b is divisible by 7}. Show that R is an equivalence relation. Find all the distinct equivalence classes of the relation R. [5 + 5]

b) (i) Solve the following system of linear equations, if possible:

$$x + y + z = 1$$
$$2x + y + 2z = 2$$
$$3x + 2y + 3z = 5$$

(ii) Find all the eigen values and the corresponding eigen vectors of the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}.$$

[5 + 5]